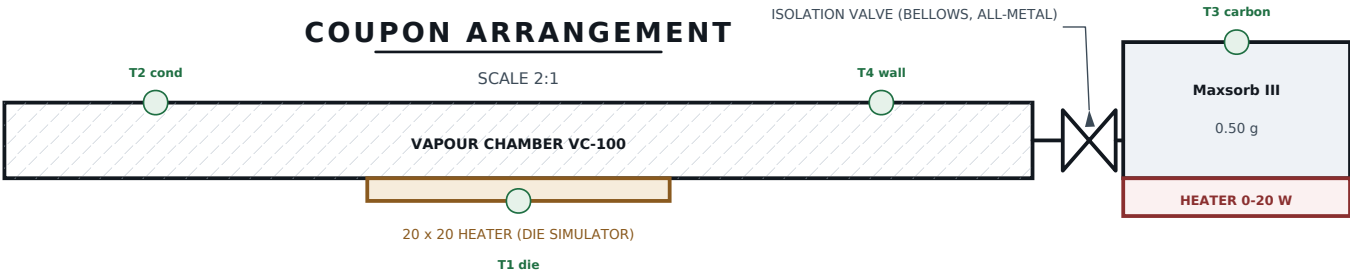


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EXPERIMENTAL COUPON — NOT A PRODUCTION CONFIGURATION — SEE PHYSICS NOTE BELOW



Mesh: 10 um SUS 316L — NOT 50 um. Maxsorb III fines pass a 50 um mesh and foul the wick.

Carbon bake-out 250 C / 8 h / < 1e-3 Pa before installation, transferred under vacuum.

Reservoir 0.5 g Maxsorb III needs >= 1.8 cm3 at 0.28 g/cc bulk density — the original 1.50 cm3

housing could not physically hold the charge. This coupon uses 30 x 12 x 6 = 2.16 cm3.

ADSORPTION EQUILIBRIUM — DUBININ-ASTAKHOV, METHANOL / MAXSORB III						
T evap	T res	P/Psat	UPTAKE	HELD	WICK FILL	RESULT
60 C	40 C	1.00	1.31 g/g	0.66 g	15 %	DRY
60 C	60 C	1.00	1.31 g/g	0.66 g	15 %	DRY
60 C	100 C	0.24	1.06 g/g	0.53 g	35 %	DRY
60 C	127 C	0.11	0.77 g/g	0.39 g	58 %	marginal
60 C	169 C	0.03	0.42 g/g	0.21 g	90 %	wet

PHYSICS NOTE — WHY THE PASSIVE VERSION CANNOT WORK

- In a sealed pure-substance chamber the pressure is set by the evaporator: $P = P_{\text{sat}}(T_{\text{evap}})$. The carbon sees the relative pressure $P / P_{\text{sat}}(T_{\text{res}})$.
- A reservoir on the condenser side always satisfies $T_{\text{res}} \leq T_{\text{evap}}$, so $P/P_{\text{sat}} \geq 1$ and the carbon saturates: it takes 0.66 g of a 0.75 g charge and the wick runs at 15 % fill. That is permanent dry-out, not pressure control.
- Keeping the wick wet needs the reservoir ~110 K ABOVE the chip. A working vapour chamber is isothermal, so no passive layout can deliver that. It requires the heater and valve drawn above.
- Overcharging to 1.90 mL keeps the wick wet, but then liquid always remains, the pressure stays at P_{sat} , and the carbon buys nothing thermally. Its only genuine residual function is as a non-condensable gas getter — for which tens of mg suffices, not 0.5 g.

COUPON vs PRODUCTION VC-100		
	WITH CARBON	WITHOUT
Wick fill at equilibrium	15 %	125 %
Q at 2 m/s forced air	0 W (dry)	113 W
Transient buffer	722 J (heater needed)	0 J
Parts count / failure modes	+ valve, heater, control	baseline

INSTRUMENTATION		
MEASUREMENT	INSTRUMENT	PURPOSE
T1..T4	type-T TC, ± 0.2 K	gradient map
Chamber pressure	capacitance manometer	verify $P_{\text{sat}}(T)$
Reservoir mass	in-situ load cell ± 1 mg	uptake vs time
Heater power	4-wire, ± 0.5 %	energy balance
Valve state	open / closed	isolate the carbon

TEST MATRIX		
#	TEST	MEASURES
1	Valve CLOSED, sweep 0-150 W	baseline R th, no carbon
2	Valve OPEN, reservoir unheated	dry-out rate + time constant
3	Valve OPEN, reservoir held 110 K above die	whether the wick recovers
4	Valve OPEN, cyclic 10 W bursts, heater regen	usable buffer energy (722 J predicted)
5	1000 h hold, valve OPEN, unheated	NCG getter benefit vs dry-out cost

COUPON NOTES

- Fluid is METHANOL for this coupon (activated carbon is hydrophobic; water uptake is poor). Accept the 7.3x lower merit number — this coupon measures adsorption, not peak cooling.
- Methanol charge 1.90 mL so the wick stays wet with the carbon saturated. Do not run at 1.00 mL.
- The valve is what makes this an experiment rather than a defect. A production unit has no valve, so it has no way to stop the carbon draining the wick.

IF TEST 4 SUCCEEDS — PATH TO A PRODUCT

- Move the reservoir onto the die's own heat path, upstream of the chamber, so it is genuinely hotter than the evaporator without a valve.
- Replace the bellows valve with a thermally actuated one-way element, or accept a control loop driving the regeneration heater.
- Budget 722 J against ~5 g of added mass and a new hermetic joint. Compare against simply adding 15 K of copper (+50 g).

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK	
TITLE		SCALE		UNITS	
MAXSORB III RESERVOIR — TEST COUPON		2:1		mm	
EXPERIMENTAL / NOT FOR PRODUCTION / INSTRUMENTED		THIRD ANGLE PROJECTION			
DRAWING No.		VC-400		REV	A
				SHEET	7 / 7
DRAWN	CHECKED	MATERIAL	FINISH	DATE	
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09	